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Universitat Pompeu Fabra Barcelona

Bachelor in Computer Engineering — Management and Information Systems

Ethical Pentesting in Virtualized Environments with EVE-NG

Feasibility Study

Volume II

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1 Feasibility Study

1.1 Initial Planning

The project has been structured in 37 tasks distributed across three main phases: Preparation (Phase 0), Avantprojecte (Phase 1), and Implementation and Final Delivery (Phases 2 and 3). Each task has been carefully defined with the following attributes:

1.1.1 Task Definition

- **Task identifier:** Unique numeric code (e.g., 0.1, 1.0, 2.1, 3.4)
- **Task name:** Descriptive title of the activity
- **Category:** Classification (Preparation, Analysis, Documentation, Development, Testing, Implementation, Milestones)
- **Estimated hours:** Time allocation based on complexity and dependencies
- **Start and end dates:** Temporal boundaries for execution
- **Completion percentage:** Current progress indicator
- **Critical path flag:** Identification of tasks that directly impact delivery deadlines

Task Categories

The 37 tasks have been distributed across 7 categories:

Category	Number of Tasks	Total Hours
Preparation	4	105h
Analysis	5	49h
Documentation	6	95h
Development	8	178h
Testing	5	123h
Implementation	6	220h
Milestones	3	95h
TOTAL	**37**	**820h**

Table 1.1: Distribution of tasks by category

Phase Breakdown

Phase 0: Preparation (July - November 2025)

- Task 0.1: HomeLab Initialization Setup (64h)
- Task 0.2: VSCode Setup for thesis development (18h)
- Task 0.3: Kali Linux Setup and Basic Tools (15h)
- Task 0.4: Home Lab As-Built Documentation (8h)

This phase established the technical infrastructure necessary for project development, including virtualization environment (EVE-NG)[1], development tools (VSCode, LaTeX)[2], [3], and pentesting platform (Kali Linux)[4]. Although this infrastructure consolidation phase extended through November, it should not be formally counted as bachelor's thesis work. The virtualization environment, development tools, and pentesting platform were pre-existing components already deployed on personal hardware. This phase primarily involved migrating and centralizing these tools onto a dedicated server infrastructure for enhanced accessibility and project organization, rather than engineering or developing new systems specifically for the bachelor's thesis.

Phase 1: Avantprojecte (October 2025 - January 2026)

- Task 1.0: Thesis Development (22h)
- Task 1.1: SMART Objectives Definition (4h)
- Task 1.2: State of the Art: Frameworks and Tools (15h)
- Task 1.3: Functional and Non-Functional Requirements (10h)
- Task 1.4: Pentesting Methodology and Validation (8h)
- Task 1.5: Design Decisions and Justification (5h)
- Task 1.6: Gantt Chart Development (20h)
- Task 1.7: Risk Analysis and Mitigation Plan (15h)
- Task 1.8: Resource Inventory (5h)

- **Milestone 1.10: Preliminary Project (40h) - Delivery: January 16, 2026**

This phase focuses on conceptual design, requirements analysis, and detailed planning. Critical tasks for the January 16 deadline include completing the pentesting methodology, risk analysis, and resource inventory.

Phase 2: Interim Report (January - March 2026)

- Task 2.1: Lab 1 Research - Reconnaissance (14h) - **CRITICAL**
- Task 2.2: Lab 2 Research - Web Vulnerabilities (14h) - **CRITICAL**
- Task 2.3: Lab 3 Research - Privilege Escalation (14h) - **CRITICAL**
- Task 2.4: Technical Write-ups Labs 1 & 2 (13h)
- Task 2.5: Topology Adjustments (30h)
- Task 2.6: Pre-validation with Pilot Users (20h) - **CRITICAL**
- Task 2.6.1: Pre-validation Revision (6h) - **CRITICAL**
- Task 2.7: Analyze and Write Hackathon Results (17h)
- Task 2.8: Interim Report Formalization (40h)
- Task 2.9: Set Up Servers, VMs and EVE-NG (20h)
- Task 2.10: Backup and Documentation of EVE-NG (10h)
- **Milestone 2.11: Intermediate Thesis (50h) - Delivery: March 20, 2026**

This phase implements the first three laboratories and validates them with pilot users. Total estimated hours: 212h.

Phase 3: Final Delivery (March - May 2026)

- Task 3.1: Lab 4 Research - Post-Exploitation (16h) - **CRITICAL**
- Task 3.2: Lab 3 Validation with Users (12h)
- Task 3.3: Technical Write-ups Labs 3 & 4 (19h)

- Task 3.4: Penetration Testing Phase 1 (Labs 1-2-3) (30h) - **CRITICAL**
- Task 3.5: Penetration Testing Phase 2 (All Labs) (25h) - **CRITICAL**
- Task 3.6: Python Automation Scripts (30h)
- Task 3.7: Final Validation with Pilot Users (18h)
- Task 3.8: Final Review and Project Polishing (25h)
- Task 3.9: Bachelor's thesis Report Writing I (15h) - **CRITICAL**
- Task 3.10: Bachelor's thesis Report Writing II (40h) - **CRITICAL**
- Task 3.11: Bachelor's thesis Report Writing III (35h) - **CRITICAL**
- Task 3.12: Final Revision (15h)
- Task 3.13: Final Report Preparation (20h)
- **Milestone 3.14: Final Report (5h) - Delivery: May 27, 2026**

This phase completes the fourth laboratory, performs exhaustive testing, develops automation scripts, and writes the final report. Total estimated hours: 315h.

1.1.2 Gantt Chart

The complete project planning has been managed using an interactive web-based Gantt chart application developed specifically for this bachelor's thesis. The tool provides:

- Visualization of all 37 tasks across the project timeline
- Identification of 13 critical path tasks
- Progress tracking through completion percentages
- Task categorization into 7 groups
- CSV import/export functionality
- High-resolution PNG export for documentation
- Version history tracking

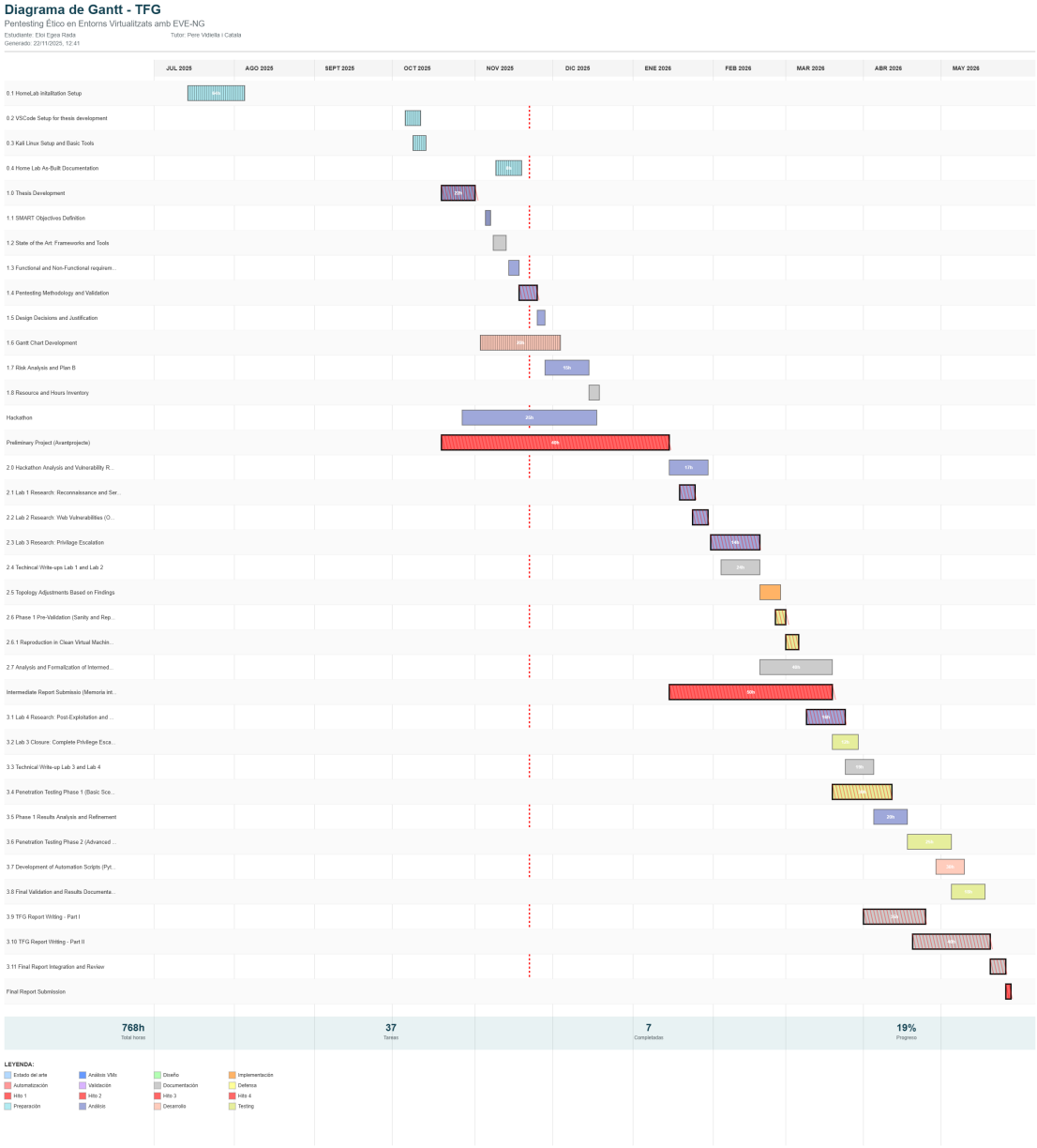

Figure 1.1: Complete Gantt chart of bachelor's thesis project planning (exported from interactive tool)

1.1.3 Critical Paths

Critical path tasks are those that directly impact the delivery deadlines and cannot be delayed without affecting subsequent milestones. The project has identified **13 critical tasks**:

ID	Critical Task	Hours	Deadline
1.4	Pentesting Methodology and Validation	8h	Nov 26, 2025
1.10	Milestone: Preliminary Project	40h	Jan 16, 2026
2.1	Lab 1 Research	14h	Jan 26, 2026
2.2	Lab 2 Research	14h	Jan 31, 2026
2.3	Lab 3 Research	14h	Feb 7, 2026
2.6	Pre-validation with Pilot Users	20h	Feb 23, 2026
2.6.1	Pre-validation Revision	6h	Mar 1, 2026
3.1	Lab 4 Research	16h	Mar 23, 2026
3.4	Penetration Testing Phase 1	30h	Apr 16, 2026
3.5	Penetration Testing Phase 2	25h	Apr 30, 2026
3.9	Bachelor's thesis Report Writing I	15h	May 1, 2026
3.10	Bachelor's thesis Report Writing II	40h	May 20, 2026
3.11	Bachelor's thesis Report Writing III	35h	May 25, 2026
TOTAL CRITICAL PATH		**277h**	

Table 1.2: Critical path tasks requiring strict adherence to deadlines

Any delay in these tasks will cascade to subsequent activities and potentially jeopardize milestone delivery dates. Special monitoring and contingency planning are allocated to these tasks.

2 Budget

2.1 Cost of Human Resources

The total estimated effort for the bachelor's thesis is **820 hours**, corresponding to the 20 ECTS credits assigned to the subject (25 hours per ECTS). The distribution is as follows:

Phase	Hours	Hourly Rate	Total Cost
Phase 0: Preparation	105h	15 €/h	1,575 €
Phase 1: Avantprojecte	188h	15 €/h	2,820 €
Phase 2: Interim Report	212h	15 €/h	3,180 €
Phase 3: Final Delivery	315h	15 €/h	4,725 €
TOTAL	**820h**		**12,300 €**

Table 2.1: Human resources cost estimation based on junior engineer hourly rate

The hourly rate of 15 €/hour corresponds to a junior cybersecurity engineer or IT student performing practical work, based on market rates in Spain for internships and entry-level positions in 2025-2026. Despite this, the cost remains theoretical, as the bachelor's thesis is not a commercial project but an academic exercise. Also, considering the educational context, this cost is not billed to any entity and less being 15 €/hour.

2.2 Cost of Hardware and Software Resources

2.2.1 Hardware Infrastructure

Resource	Quantity	Unit Cost	Total
HomeLab Server (CPU 28-core, 32GB RAM, +32TB SSD)	1	800 €	800 €
Development Workstation (Laptop + Desktop Computer)	1	1200 €	1200 €
Network Equipment (Router, Switch)	1 set	150 €	150 €
External Storage (Backup)	1	100 €	100 €
SUBTOTAL Hardware			**2,250 €**

Table 2.2: Hardware infrastructure costs

Note: The HomeLab server is a one-time investment that will be reused for future projects and courses, amortizing its cost over multiple years. The development workstation and network equipment are also reusable assets. Although that the use of the HomeLab server, a laptop and a desktop computer is already assumed on Phase 0, and its cost is not directly billed to the project, it's included here for completeness.

Software and Licenses

Resource	License Type	Unit Cost	Total
EVE-NG Community Edition	Free/Open Source	0 €	0 €
Kali Linux	Free/Open Source	0 €	0 €
VMware Workstation Pro (Educational)	Educational License	0 €	0 €
Metasploitable VMs	Free/Open Source	0 €	0 €
DVWA (Damn Vulnerable Web App)	Free/Open Source	0 €	0 €
Visual Studio Code	Free/Open Source	0 €	0 €
LaTeX (TeX Live)	Free/Open Source	0 €	0 €
Git + GitHub (Student Pack)	Free/Student	0 €	0 €
Burp Suite Community	Free	0 €	0 €
Wireshark	Free/Open Source	0 €	0 €
SUBTOTAL Software			**0 €**

Table 2.3: Software and license costs (all free/open-source for educational use)

A significant advantage of this project is the reliance on free and open-source tools, which reduces costs to zero for software licensing while maintaining professional-grade capabilities.

2.3 Other Costs

Concept	Cost
Electricity consumption (820h × 0.15 kWh × 0.20 €/kWh)	25 €
Internet connectivity (9 months × 40 €/month)	360 €
Documentation printing and binding (3 copies)	60 €
Contingency fund (5% of total)	635 €
TOTAL Other Costs	**1,080 €**

Table 2.4: Additional operational costs

2.4 Total Budget Summary

Category	Total Cost
Human Resources	12,300 €
Hardware Resources	2,250 €
Software Resources	0 €
Other Costs	1,080 €
TOTAL PROJECT BUDGET	**15,630 €**

Table 2.5: Complete project budget breakdown

3 Feasibility Analysis

3.1 Technical Feasibility

3.1.1 Infrastructure Availability

The project demonstrates **high technical feasibility** due to the following factors:

- **Completed Infrastructure Setup**: Phase 0 (Preparation) has been 100% completed, with HomeLab server, EVE-NG platform, Kali Linux, and development environment fully operational and documented.
- **Open-Source Ecosystem**: All software components (EVE-NG, Kali Linux, vulnerable VMs, pentesting tools) are mature, well-documented, and widely adopted in professional cybersecurity training.[5]
- **Proven Technology Stack**: The selected technologies (VMware/EVE-NG for virtualization, Python/Bash for scripting, LaTeX for documentation) are industry-standard and have established best practices.[6]
- **Academic Support**: Access to institutional resources at Tecnocampus, including academic licenses for software (VMware Workstation Pro), library resources for research, and tutor expertise in cybersecurity and network administration.

3.1.2 Technical Competencies

The project requires competencies in multiple areas:

Area	Required Competency	Level
Network Administration	Configuration of virtual networks, VLANs, routing, firewall rules	Intermediate
Operating Systems	Linux administration, Windows Server configuration	Intermediate
Virtualization	EVE-NG topology design, VM management, resource optimization	Advanced
Pentesting	Reconnaissance, exploitation, privilege escalation, web vulnerabilities	Intermediate
Scripting	Python and Bash for automation and validation	Intermediate
Documentation	LaTeX, technical writing, academic formatting	Intermediate

Table 3.1: Required technical competencies and current proficiency level

The student has demonstrated competence in these areas through completed courses (Network Administration, Operating Systems, Software Engineering) and the successful completion of Phase 0 infrastructure setup.

3.1.3 Risk Assessment Framework

Following the five-phase risk assessment methodology defined in NIST SP 800-30, this section systematically identifies, evaluates, and prioritizes technical risks to ensure appropriate mitigation strategies.

Phase 1: Asset Identification and Scope Definition

The project's critical assets have been identified and categorized according to the CIA triad (Confidentiality, Integrity, Availability):

Hardware Assets:

- HomeLab server (28-core CPU, 32GB RAM, 32TB SSD) - Avail-

ability critical

- Development workstations (laptop + desktop) - Integrity critical
- Network equipment (router, switch) - Availability critical
- External storage devices (backup) - Confidentiality + Availability critical

Software Assets:

- EVE-NG community edition - Availability critical
- Kali Linux distribution - Integrity critical
- Vulnerable VMs (Metasploitable, DVWA) - Integrity critical
- Development tools (VSCode, Git, LaTeX) - Integrity critical

Data Assets:

- Thesis source code and documentation - Confidentiality + Integrity critical
- Lab topology configurations - Integrity + Availability critical
- Student learning materials - Availability critical
- Project research findings - Confidentiality + Integrity critical

Infrastructure Assets:

- Virtual network topology (EVE-NG) - Availability critical
- Network segmentation (VLANs) - Integrity critical
- Backup systems - Availability critical

Scope Definition: The risk assessment focuses on infrastructure and technical risks within the controlled laboratory environment. Excluded from scope: organizational risks (staffing, budget), regulatory risks (compliance), and external threats beyond the academic context.

Phase 2: Threat and Vulnerability Identification

Threat Identification:

Threats are categorized by source according to NIST SP 800-30[7]:

- **Hardware Threats:** Disk failures, power supply degradation, CPU overheating, memory exhaustion
- **Software/Configuration Threats:** Outdated VM images, incompatible software versions, configuration drift, EVE-NG platform crashes
- **Knowledge Gap Threats:** Insufficient expertise in advanced networking, virtualization optimization, security best practices
- **Resource Constraint Threats:** Limited RAM for simultaneous VM operations, storage capacity limits, bandwidth limitations
- **Integration Threats:** EVE-NG network compatibility issues with host systems, tool interoperability problems
- **Data Loss Threats:** Accidental deletion, corruption, hardware failure causing data unavailability

Vulnerability Identification:

Vulnerabilities are the weaknesses that allow threats to impact assets:

- **Hardware Vulnerabilities:**
  - Aging server components (3-5 year lifespan approaching end-of-life)
  - Single point of failure in HomeLab infrastructure (no redundancy)
  - Limited RAM capacity for production-scale virtualization
- **Software Vulnerabilities:**
  - EVE-NG community edition limitations (no commercial support)
  - Outdated VM image libraries

  - Dependency on end-of-life operating system versions

- **Operational Vulnerabilities:**
  - Inadequate documentation of network topology state
  - Limited backup redundancy (single backup location)
  - No automated disaster recovery procedures

- **Competency Vulnerabilities:**
  - Limited hands-on experience with large-scale virtualization
  - Advanced network configuration knowledge gaps
  - Insufficient production-grade infrastructure management experience

Phase 3: Impact and Likelihood Analysis

Risk assessment requires evaluating both likelihood (probability) and impact (consequence) of each identified threat exploiting a vulnerability. Scales follow NIST SP 800-30:

Probability Scale:

- **Low:** < 25% probability within project timeline
- **Medium:** 25-75% probability
- **High:** > 75% probability

Impact Scale:

- **Low:** Minor delays, easy workarounds possible
- **Medium:** Project delays, workarounds required
- **High:** Project jeopardy, timeline at significant risk

Risk Assessment Matrix:

Risk Scenario	Probability	Impact	Risk Level	Primary Asset
EVE-NG Performance Issues	Medium	High	HIGH	Infrastructure, Labs
VM Image Incompatibility	Medium	Medium	MEDIUM	Development, Labs
Knowledge Gaps in Advanced Networking	Medium	Medium	MEDIUM	Development, Timeline
Hardware Degradation / Disk Failure	Low	High	MEDIUM	Infrastructure
Data Loss (insufficient backup)	Low	High	MEDIUM	Documentation
Configuration Documentation Gaps	Medium	Medium	MEDIUM	Operations

Table 3.2: Risk assessment matrix with probability and impact evaluation (NIST SP 800-30)

Phase 4: Risk Evaluation and Prioritization

Based on the impact-likelihood analysis, risks are prioritized to guide mitigation investment:

HIGH Priority Risks:

1. **EVE-NG Performance Issues** - Infrastructure bottlenecks causing lab functionality failures. Medium probability, high impact. Requires immediate resource optimization and monitoring implementation.

MEDIUM Priority Risks:

1. **VM Incompatibility** - Disparate image formats and versions causing integration problems. Medium probability, medium impact.
2. **Knowledge Gaps** - Team experience gaps in advanced networking and virtualization. Medium probability, medium impact on timeline.
3. **Configuration Documentation** - Inadequate topology documentation complicating troubleshooting and knowledge transfer. Medium probability, medium impact.
4. **Data Loss** - Insufficient backup redundancy creating single point of failure. Low probability, high impact if occurs.

LOW Priority Risks:

1. **Hardware Degradation** - Aging server components approaching end-of-life. Low probability, manageable through preventive maintenance.

Phase 5: Mitigation Strategies

For each identified risk, mitigation strategies follow NIST SP 800-30 response approaches:

HIGH Priority Risk Mitigation:

- **EVE-NG Performance Issues** [REDUCTION Strategy]:
  - Hardware dimensioning: Allocate 32GB RAM, 8-core CPU dedicated to EVE-NG to prevent resource exhaustion
  - Resource optimization: Implement memory tuning, network simulation limits, VM suspension when idle
  - Monitoring implementation: Deploy Prometheus + Grafana (Phase 2) for real-time performance alerting
  - Documented fallback: Maintain VirtualBox installation as alternative hypervisor if critical failure occurs
  - Result: Reduces probability from Medium to Low

MEDIUM Priority Risk Mitigation:

- **VM Incompatibility** [PREVENTION Strategy]:
  - Early testing: Validate all VM images (Metasploitable, DVWA, Kali Linux)[8] in target environment before production use
  - Format standardization: Standardize on QCOW2 for EVE-NG compatibility
  - Image redundancy: Maintain alternative repositories (VulnHub, GitHub) for rapid replacement
  - Result: Reduces probability from Medium to Low
- **Knowledge Gaps** [PREVENTION + REDUCTION]:

  - Structured learning: Task 1.2 (State of the Art: 15h) provides comprehensive research on frameworks and tools
  - Continuous upskilling: Learning during implementation phases (Labs 1-4 research tasks)
  - Expert consultation: Access to tutor expertise in cybersecurity and network administration
  - Technical references: NIST SP 800-115[9], OWASP Testing Guide[10]
  - Result: Reduces impact from Medium to Low-Medium
- **Configuration Documentation** [PREVENTION]:
  - Continuous documentation: Update topology diagrams during development
  - Automated export: Use EVE-NG built-in topology export functionality
  - Version control: All network configs via Git repository
  - Enforcement: All topology changes require corresponding documentation updates before milestone completion
  - Result: Reduces probability from Medium to Low
- **Data Loss** [REDUCTION]:
  - Daily automated backups: Encrypt and backup critical files to external storage device
  - Version control: All source code and documentation in Git repository with commit history
  - Off-site backup: Cloud storage (student account) with encryption for geographic redundancy
  - Restore testing: Quarterly restore tests to verify backup integrity
  - Result: Reduces both probability and impact (Low probability, Low impact)

LOW Priority Risk Mitigation:

- **Hardware Degradation** [PREVENTION]:

- Preventive maintenance: Quarterly hardware diagnostics (disk health, temperature monitoring)
- Component monitoring: Automated alerts for disk S.M.A.R.T. errors, CPU temperature thresholds
- Replacement planning: Component procurement plan with 6-month horizon for critical parts
- Result: Enables early detection and prevents critical failure

Risk Assessment Conclusion:

All identified risks have been systematically evaluated and assigned appropriate mitigation strategies. No individual risk is considered project-blocking. The comprehensive mitigation plan following NIST SP 800-30 methodology ensures the project can proceed with acceptable technical risk levels. The infrastructure is well-prepared, technology stack is appropriate, and mitigation strategies are concrete and implementable. Main technical challenges are manageable with proper planning and documented contingency measures.

4 Economic Feasibility

4.1 Cost-Benefit Analysis

The project presents **excellent economic feasibility** for the following reasons:

- **Zero Software Licensing Costs:** The exclusive use of free and open-source software eliminates recurring licensing expenses, making the solution sustainable for long-term educational use.
- **Low Infrastructure Investment:** The hardware infrastructure (2,250 €) is a one-time investment that can be reused across multiple academic years and courses, amortizing costs over 3-5 years.
- **Reusability and Scalability:** The teaching package is designed to be reused in multiple course editions (Introduction to Cybersecurity) with minimal maintenance costs, generating value beyond the initial development.
- **Institutional Value:** The deliverable provides long-term value to Tecnocampus by establishing a reproducible, automated laboratory infrastructure that reduces preparation time for faculty and enhances learning outcomes for students.

4.2 Cost Comparison with Alternatives

Alternative	Description	Annual Cost
This Project	Custom EVE-NG labs with automation, tailored to curriculum	0 €/year*
HackTheBox Academy	Commercial platform with pre-built labs and guided exercises	1,200 €/year
TryHackMe Education	Gamified learning platform with structured learning paths	800 €/year
Cybrary for Teams	Video-based training with labs and certification prep	1,500 €/year
Custom Enterprise Labs	Outsourced lab development by specialized company	8,000–15,000 €

Table 4.1: Cost comparison with commercial alternatives (*excludes initial development)

While commercial platforms offer convenience, they lack customization to specific curriculum needs, do not integrate with institutional learning management systems, and incur recurring annual costs. This project provides a tailored, cost-effective, and sustainable solution.

Conclusion: The project demonstrates **high economic feasibility** with minimal ongoing costs, significant institutional value, and favorable comparison to commercial alternatives.

5 Environmental Feasibility

5.1 Energy Consumption Analysis

The environmental impact of this project is primarily related to energy consumption during development and operation:

- **Development Phase (820 hours):**
  - HomeLab server power consumption: 150W average
  - Development workstation: 65W average
  - Network equipment: 15W
  - Total power: 230W × 820h = 188.6 kWh
  - CO^2 emissions (Spain avg. 0.25 kg CO^2/kWh): 47.15 kg CO^2[11]
  - **Solar Energy Offset (Home Installation):** Home infrastructure powered by 5kWp rooftop solar installation:
    * 2025 annual generation: 6,634.9 kWh (real data from Can Regasol installation)
    * 2025 household consumption: 7,708.9 kWh (grid + solar)
    * Daily average generation during development: ~18.2 kWh/day
    * Estimated solar offset for development phase: 34% of 188.6 kWh (~64.1 kWh from solar)
    * Adjusted CO^2 emissions (with solar): 31.95 kg CO^2 (32% reduction)[11], [12]
- **Operational Phase (per academic year):**
  - Estimated usage: 4 laboratory sessions × 30 students × 3 hours = 360 hours/year
  - Power consumption: 230W × 360h = 82.8 kWh/year
  - CO^2 emissions: 20.7 kg CO^2/year

5.2 Comparison with Traditional Alternatives

Scenario	Description	Annual CO_2	With Solar*
This Project	Virtualized labs on shared HomeLab infrastructure	20.7 kg	14.1 kg
This Project (Dev)	Development phase with solar offset (6.6 MWh/year avg)	47.15 kg	31.95 kg
Individual VMs	Each student runs labs on personal computer (30 × 200W × 12h)	180 kg	–
Cloud-Based Labs	AWS/Azure hosted environments (datacenter PUE 1.5)	95 kg	–
Physical Network Lab	Dedicated hardware devices (switches, routers, servers)	450 kg	–

Table 5.1: CO^2 emissions comparison across laboratory approaches (*Solar offset based on 5kWp installation producing 6.6 MWh/year)

5.3 Sustainability Measures

The project incorporates several sustainability practices:

- **Renewable Energy Integration (Solar-Powered Infrastructure):** The entire HomeLab and development environment operates on a 5kWp rooftop photovoltaic installation located at Can Regasol (Catalunya, Spain). Real operational data from 2025 demonstrates 6,634.9 kWh annual generation with household consumption of 7,708.9 kWh, indicating robust solar capacity with 3,796.4 kWh annual export to grid. This configuration offsets approximately 34% of development phase energy consumption (64.1 kWh from 188.6 kWh total), reducing CO^2 emissions by 32% compared to grid-only scenarios. The installation demonstrates practical implementation of distributed renewable energy for academic research infrastructure[11], [12].

- **Resource Optimization:** VMs are configured with minimal resource allocation, suspended when idle, and share infrastructure efficiently through virtualization.

- **Digital Documentation:** All documentation is maintained in digital format (LaTeX source, PDF outputs), avoiding paper waste except for required printed thesis copies (3 units).
- **Hardware Longevity:** The HomeLab server is designed for longevity (3-5 years) with modular components that can be upgraded rather than replaced entirely.
- **Open-Source Philosophy:** Reliance on open-source software reduces electronic waste by extending the lifespan of existing hardware that might be considered obsolete for commercial software requirements.

Conclusion: The project presents **good environmental feasibility** with significantly lower carbon footprint compared to physical lab alternatives or individual VM deployments. The virtualized approach maximizes resource efficiency and minimizes environmental impact.

6 Legal Aspects

6.1 Intellectual Property and Licensing

The project complies with intellectual property regulations:

- **Open-Source Software:** All utilized software (EVE-NG, Kali Linux, Metasploitable, DVWA) is licensed under permissive open-source licenses (GPL, MIT, Apache 2.0) that explicitly allow educational use, modification, and redistribution.
- **Virtual Machine Images:** Vulnerable VMs (Metasploitable, DVWA) are provided by their respective projects with explicit permission for cybersecurity training and research purposes.
- **Documentation Ownership:** Per UPF regulations (Article 10), intellectual property of the bachelor's thesis belongs to the student. Exploitation rights default to the student unless otherwise agreed with the tutor and institution.
- **Third-Party Content:** All references, citations, and external resources used in documentation follow IEEE citation standards, ensuring proper attribution and avoiding plagiarism.

6.2 Ethical and Legal Considerations for Pentesting

The project adheres to ethical hacking principles and legal boundaries:

- **Controlled Environment:** All pentesting activities are conducted within isolated virtualized environments (EVE-NG sandbox), with no connection to production networks or systems outside the lab.
- **Authorized Targets:** Only intentionally vulnerable systems (Metasploitable, DVWA) designed explicitly for pentesting training are used as targets[8].

- **Educational Purpose:** The project is conducted under academic supervision (Pere Vidiella i Catalan, tutor) within the scope of the "Introduction to Cybersecurity" course curriculum, satisfying the educational exception for security research.
- **Data Privacy:** No personally identifiable information (PII) or sensitive data is processed during laboratory exercises. All scenarios use synthetic data or publicly available test datasets.
- **Responsible Disclosure:** In the unlikely event that previously unknown vulnerabilities are discovered in open-source software during testing, they will be reported responsibly to the respective maintainers following coordinated disclosure practices.

6.3 Compliance with Academic Regulations

The project follows Tecnocampus ESUPT bachelor's thesis regulations (approved September 30, 2021):

- **Plagiarism Prevention:** All work is original, properly cited, and verified using plagiarism detection tools. Adherence to Article 11 (Plagiarism) ensuring academic integrity.
- **Evaluation Criteria:** The project structure follows Chapter 9 (Evaluation) requirements, including continuous assessment (avant-projecte, interim report, final delivery) and tribunal defense.
- **Documentation Standards:** Formatting adheres to the "Guia per a la redacció i edició del bachelor's thesis" (ESUPT guidelines), ensuring compliance with submission requirements.

Conclusion: The project demonstrates **full legal compliance** with intellectual property regulations, ethical hacking principles, and institutional academic standards.

7 Gender and Diversity Perspective

7.1 Inclusive Language and Representation

The project incorporates gender and diversity considerations:

- **Inclusive Language:** All documentation (thesis, technical write-ups, student guides) uses gender-neutral language where possible, avoiding gender-biased assumptions.
- **Diverse Examples:** Laboratory scenarios and fictional user accounts employ diverse names and backgrounds, reflecting the multicultural and gender-diverse nature of the cybersecurity profession.
- **Accessibility:** Documentation is structured to be accessible to students with diverse learning needs, including clear formatting, descriptive headings, and optional use of dyslexia-friendly fonts (OpenDyslexic) in LaTeX template.

7.2 Addressing the Gender Gap in Cybersecurity

Cybersecurity remains a male-dominated field. This project contributes to bridging the gender gap:

- **Accessible Entry Point:** Well-documented, structured laboratories lower barriers to entry for students traditionally underrepresented in technical fields, providing a supportive learning environment.
- **Ethical Framework:** Emphasizing the ethical aspects of cybersecurity (responsible disclosure, defensive security, societal impact) may appeal to a broader demographic beyond the stereotype of "hacker culture."
- **Collaborative Learning:** The design encourages collaborative problem-solving and knowledge sharing rather than competitive dynamics, fostering an inclusive learning culture.

Conclusion: The project demonstrates **awareness and commitment** to gender and diversity considerations, incorporating inclusive language, diverse representation, and accessibility measures that align with modern educational best practices.

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Bibliography Notes

This bibliography compiles all sources referenced throughout this feasibility analysis. References are organized using the IEEE citation style, which is the standard for computer science and engineering publications. The complete reference database is maintained in `references.bib`.

Reference Organization

The bibliography includes references across the following categories:

- **Infrastructure & Virtualization:** EVE-NG, GNS3, Proxmox, VMware
- **Pentesting Tools:** Kali Linux, Metasploit, Burp Suite
- **Educational Platforms:** HackTheBox, TryHackMe, SEED Labs, VulnHub
- **Security Frameworks:** NIST, OWASP, CEH, PTES
- **Institutional:** Tecnocampus ESUPT normativa, GEISI curriculum
- **Environmental:** Green IT research, CO^2 emissions data
- **Cybersecurity Education:** Hands-on learning research, curriculum design